LAB MANUAL

Subject :- Object Oriented Programming (BE04000231)

Branch :- CE, IT & CSE (DS)

Collage :- SAL Collage of Engineering

INDEX

No.	Title	Date of practical	Date of submission	Sign
1.	**Introduction to java.** 1. Write a program to display "Welcome To Java World". 2. Write a program to find whether the number is prime or not. 3. Write a program to find a greater number among given three numbers using a) ternary operator and b)nested if. 4. Write a program to print the Fibonacci series.			
2.	**Introduction to Methods and Arrays** 1. Write a method to enter two integers and compute the gcd of two integers. 2. Write a test program that prompts the user to enter ten numbers, invoke a method to reverse the numbers, display the numbers.			
3.	**Introduction to class, object and constructor** **1.** Write a program that declares a class named Person. It should have instance variables to record name, age and salary. Use new operator to create a Person object. Set and display its instance variables. Add a constructor to the Person class developed above. **2.** The employee list for a company contains employee code, name, designation and basic pay. The employee is given HRA of 10% of the basic and DA of 45% of the basic pay. The total pay of the employee is calculated as Basic pay+HRA+ DA. Write a class to define the details of the employee. Write a constructor to assign the required initial values. Add a method to calculate HRA, DA and Total pay and print them out. Write another class with a main method. Create objects for three			

	different employees and calculate the HRA, DA and total pay.			
4.	**String class, StringBuilder Class and StringBuffer class.** 1. Write a Java Program to Count Number of Words in Given String. 2. Write a Java Program to Count the Occurrences of Each Character in String.			
5.	**Inheritance and Polymorphism** 1. Write a program to illustrate the concept of class with Method overloading. 2. Write a program to illustrate the concept of Single inheritance. 3. Write a program to illustrate the concept of Multi level inheritance			
6.	**How to use Exception handling** 1. Write an exception class for a time of day that can accept only 24 hour representation of clock hours. Write a java program to input various formats of timings and throw suitable error messages.			
7.	**To Understand concept of abstract class.** 1. The abstract Vegetable class has three subclasses named Potato, Brinjal and Tomato. Write an application that demonstrates how to establish this class hierarchy. Declare one instance variable of type String that indicates the color of a vegetable. Create and display instances of these objects. Override the toString() method of Object to return a string with the name of the vegetable and its color. 2. Create an abstract class 'Shape' with two members base and height , a member function for initialization and a pure			

	virtual function to compute area(). Derive two specific classes 'Triangle' and 'Rectangle' which override the function area(). Use these classes in main function and display the areaof triangle and rectangle.			
8.	**To Understand concept of interface.** 1. Write a program that illustrates interface inheritance. Interface A is extended by A1 and A2. Interface A12 inherits from both P1 and P2. Each interface declares one constant and one method. Class B implements A12.Instantiate B and invoke each of its methods. Each method displays one of the constants.			
9.	**To Understand concept JavaFX and UI controls.** 1. Write a program that displays an image in a label, a title in a label, and text in the text area as follows: 			
10.	**To Understand concept of binary I/O, recursion and generics** 1. Write a program to count the total number of chars, words, lines, alphabets, digits, white spaces in givenfile. 2. Write the program for factorial using the concept of recursion.			

EXPERIMENT : 1

Title : Introduction to java

AIM : Write a program to display “Welcome To Java World”.

Code :

```
public class hello_world {
    public static void main (String args[])
    {
    System.out.println("hello world");
    System.out.println("Kartik patel ,enrolment no :-
201130116009");
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac hello_world.java } ; if ($?) { java hello_world }
hello world
Kartik patel ,enrolment no :- 201130116009
PS D:\kartik_java_prac>
```

AIM : Write a program to find whether the number is prime or not.

Code :

```
import java.util.*;
class prime {
    public static void main(String args[]) {
        Scanner input = new Scanner(System.in);
        int i, num, flag = 0, m;
        System.out.println("Enrollment no :-
201130116009 ");
        System.out.print("Enter the number : ");
        num = input.nextInt();
        if (num == 1) {
            System.out.println("1 is prime number");
        }
        for (i = 2; i < num; i++) {
            if (num % i == 0) {
                flag = 0;
                break;
            } else {
                flag = 1;
            }
        }
```

```
            if (flag == 1) {
                System.out.println("Prime number");
            } else {
                System.out.println("Not Prime number");
            }
        }
    }
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac prime.java } ; if ($?) { java prime }
Enrollment no :- 201130116009
Enter the number : 11
Prime number
PS D:\kartik_java_prac>
```

AIM : Write a program to find a greater number among given three numbers using a) ternary operator and b)nestedif.

Code : 1 . ternary operator

```
import java.util.Scanner;
public class ternary {
    public static void main(String[] args) {
        int a, b, c, d;
```

```
        Scanner s = new Scanner(System.in);
        System.out.println("Enrollment no :-
201130116009 ");
        System.out.println("Enter all three numbers:");
        a = s.nextInt();
        b = s.nextInt();
        c = s.nextInt();
        d = c > (a > b ? a : b) ? c : ((a > b) ? a : b);
        System.out.println("Largest Number:" + d);
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac ternary.java } ; if ($?) { java ternary }
Enrollment no :- 201130116009
Enter all three numbers:
25
50
75
Largest Number:75
PS D:\kartik_java_prac>
```

Code : 2. Nested if

```
import java.util.*;
```

```
public class ternary {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        double n1 ,n2, n3;
        System.out.println("Enrollment no :-
201130116009 ");
        System.out.println("enter 1st number ");
        n1 = input.nextInt();
        System.out.println("enter 2nd number ");
        n2 = input.nextInt();
        System.out.println("enter 3rd number ");
        n3 = input.nextInt();
        if( n1 >= n2 && n1 >= n3)
            System.out.println(n1 + " is the largest
number.");

        else if (n2 >= n1 && n2 >= n3)
            System.out.println(n2 + " is the largest
number.");
```

```
        else
            System.out.println(n3 + " is the largest number.");
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac ternary.java } ; if ($?) { java ternary }
Enrollment no :- 201130116009
enter 1st number
20
enter 2nd number
40
enter 3rd number
60
60.0 is the largest number.
PS D:\kartik_java_prac>
```

AIM : Write a program to print the Fibonacci series

Code : Fibonacci series

```
import java.util.*;

public class fibonacci {
    public static void main(String args[]) {
```

```
        int n1 = 0, n2 = 1, i, n, n3;
        Scanner input = new Scanner(System.in);
        System.out.println("Enrollment no :-
201130116009 ");
        System.out.println("Enter the number");
        n  = input.nextInt();
        System.out.print(n1);
        System.out.print(" " + n2);
        for (i = 0; i < n; i++) {
            n3 = n1 + n2;
            System.out.print(" " + n3);

            n1 = n2;
            n2 = n3;
        }
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac fibonacci.java } ; if ($?) { java fibonacci }
Enrollment no :- 201130116009
Enter the number
15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987
PS D:\kartik_java_prac>
```

EXPERIMENT NO: 2

Title :- Introduction to Methods and Arrays

AIM : Write a method to enter two integers and compute the GCD of two integers.

Code :

```
import java.util.Scanner;


public class gcd {
   public static void main(String[] args) {
      int a, b, gcd = 0;
      Scanner sc = new Scanner(System.in);
      System.out.println("enrollment no :-
201130116009");
      System.out.print("Enter the First Number: ");
      a = sc.nextInt();
      System.out.print("Enter the Second Number: ");
      b = sc.nextInt();
      gcd = findGCD(a, b);
      System.out.println("GCD of " + a + " and " + b + " =
" + gcd);
```

```
    }

    public static int findGCD(int a, int b) {
        while (b != 0) {
            if (a > b) {
                a = a - b;
            } else {
                b = b - a;
            }
        }
        return a;
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac gcd.java } ; if ($?) { java gcd }
enrollment no :- 201130116009
Enter the First Number: 25
Enter the Second Number: 50
GCD of 25 and 50 =  25
PS D:\kartik_java_prac>
```

AIM : Write a test program that prompts the user to enter ten numbers, invoke a method to reverse the numbers, display the numbers.

Code :-

```
class reverse {
   public static void main(String args[]) {
      System.out.println("Enrollment no
201130116009");
      int a[] = { 26, 35, 19, 45, 11 };
      for (int i = 0; i < a.length; i++)
         System.out.println(a[i]);
      System.out.println("the array is written in reverse
order");
      for (int i = a.length - 1; i >= 0; i--)
         System.out.println(a[i]);
   }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac reverse.java } ; if ($?) { java reverse }
Enrollment no 201130116009
26
35
19
45
11
the array is written in reverse order
11
45
19
35
26
PS D:\kartik_java_prac>
```

EXPERIMENT : 3

Title :- Introduction to class, object and constructor

AIM : Write a program that declares a class named Person. It should have instance variables to record name, age and salary. Use new operator to create a Person object. Set and display its instance variables. Add a constructor to the Person class developed above.

Code :

```
import java.util.*;

class person {
    String name;
    int age;
    int salary;

    person(String n, int a, int s) {
        name = n;
        age = a;
        salary = s;
```

```
    }

    void display() {
        System.out.println("");
        System.out.println(name);
        System.out.println(salary);
        System.out.println(age);
    }
}

class persons {
    public static void main(String args[]) {
        System.out.println("Enrollment no :-
201130116009");

        Scanner input = new Scanner(System.in);

        System.out.println("enter name :");
        String name = input.next();
        System.out.println("enter salary :");
```

```
        int salary = input.nextInt();
        System.out.println("enter age :");
        int age = input.nextInt();
        person obj = new person(name, age, salary);
        obj.display();
    }
}
```

output :-

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac persons.java } ; if ($?) { java persons }
Enrollment no :- 201130116009
enter name :
kartik
enter salary :
25000
enter age :
18

kartik
25000
18
PS D:\kartik_java_prac>
```

AIM : The employee list for a company contains employee code, name, designation and basic pay. The employee is

given HRA of 10% of the basic and DA of 45% of the basic pay. The total pay of the employee is calculated as Basic pay + HRA + DA. Write a class to define the details of the employee. Write a constructor to assign the required initial values. Add a method to calculate HRA, DA and Total pay and print them out. Write another class with a main method. Create objects for three different employees and calculate the HRA, DA and total pay.

Code :

```
import java.util.*;


class detail {
    double ep_code;
    String name;
    String designation;
    double basic, total;

    detail(double e, String n, String d, double b) {
        ep_code = e;
        name = n;
        designation = d;
        basic = b;
```

```
    }

    void cal() {
        double hra, da;
        hra = basic * (0.1);
        da = basic * (0.45);
        total = basic + hra + da;
    }

    void display() {
        System.out.println("Employee Code : " + ep_code);
        System.out.println("Employee name: " + name);
        System.out.println("Employee designation: " +
designation);
        System.out.println("Employee Total salary:" +
total);
    }
}

public class employee {
    public static void main(String args[]) {
```

```
        System.out.println("enrolment no :-
201130116009");
        for (int i = 0; i < 3; i++) {
            Scanner input = new Scanner(System.in);
            System.out.println("Enter the employee Code :
");
            double ep_code = input.nextDouble();
            System.out.println("Enter the employee
name:");
            String name = input.next();
            System.out.println("Enter the employee
designation:");
            String designation = input.next();
            System.out.println("Enter the employee Basic
pay:");
            double basic = input.nextDouble();
            detail obj = new detail(ep_code, name,
designation, basic);
            obj.cal();
            obj.display();
        }
    }
```

```
}
```

Output :

```
24
Enter the employee name:
KARTIK
Enter the employee designation:
TL
Enter the employee Basic pay:
25000
Employee code: 24.0
Employee name: KARTIK
Employee designation: TL
Employee Total salary:38750.0
Enter the employee code:
9
Enter the employee name:
AKSHIT
Enter the employee designation:
ATL
Enter the employee Basic pay:
20000
Employee code: 9.0
Employee name: AKSHIT
Employee designation: ATL
Employee Total salary:31000.0
Enter the employee code:
23
Enter the employee name:
VANSHI
Enter the employee designation:
INTERN
Enter the employee Basic pay:
15000
Employee code: 23.0
Employee name: VANSHI
Employee designation: INTERN
Employee Total salary:23250.0
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac ternary.java } ; if ($?) { java ternary }
```

Experiment : 4

Title :- String class, StringBuilder Class and StringBuffer class.

AIM : Write a Java Program to Count Number of Words in Given String

Code :

```
public class count1{


   public static void main(String[] args) {
         System.out.println("Enrollment no :-
201130116009");
         String str = "i love india";
         int count = 1;
         for (int i = 0; i < str.length() - 1; i++)
         {
            if((str.charAt(i) == ' ') && (str.charAt(i + 1) != '
'))
            {
               count++;
            }
```

```
            }
            System.out.println("Number of words in a string : " + count);
        }
    }
```

Output : -

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac count1.java } ; if ($?) { java count1 }
Enrollment no :- 201130116009
Number of words in a string : 3
PS D:\kartik_java_prac>
```

AIM : - Write a Java Program to Count the Occurrences of Each Character in String

Code : -

```
public class countch {
    static final int MAX_CHAR = 256;

    static void getOccuringChar(String str) {
```

```
        int count[] = new int[MAX_CHAR];
        int len = str.length();
        for (int i = 0; i < len; i++)
            count[str.charAt(i)]++;
        char ch[] = new char[str.length()];
        for (int i = 0; i < len; i++) {
            ch[i] = str.charAt(i);
            int find = 0;
            for (int j = 0; j <= i; j++) {
                if (str.charAt(i) == ch[j])
                    find++;
            }
            if (find == 1)
                System.out.println("The occurrence of " +
str.charAt(i) + " is: " + count[str.charAt(i)]);
        }
    }

    public static void main(String args[]) {
        System.out.println("enrollment no :-
201130116009");
```

```
        String str = "PATEL KARTIK KALPESHKUMAR";
        System.out.println("string is " + str);
        getOccuringChar(str);
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac countch.java } ; if ($?) { java countch }
enrollment no :- 201130116009
string is PATEL KARTIK KALPESHKUMAR
The occurrence of P is: 2
The occurrence of A is: 4
The occurrence of T is: 2
The occurrence of E is: 2
The occurrence of L is: 2
The occurrence of   is: 2
The occurrence of K is: 4
The occurrence of R is: 2
The occurrence of I is: 1
The occurrence of S is: 1
The occurrence of H is: 1
The occurrence of U is: 1
The occurrence of M is: 1
PS D:\kartik_java_prac>
```

Experiment-5

Inheritance and Polymorphism

AIM : Write a program to illustrate the concept of class with Method overloading.

Code :-

```
public class overloading {
    public static void main(String args[]) {
        System.out.println("enrolment no :-
201130116009");
        System.out.println("Kartik patel");
        System.out.println("Sum of two integers   ");
        Sum(10, 20);
        System.out.println("Sum of two double numbers
");
        Sum(10.5, 20.4);
        System.out.println("Sum of three integers ");
        Sum(10, 20, 30);
    }


    public static void Sum(int num1, int num2) {
```

```
        int ans;
        ans = num1 + num2;
        System.out.println(ans);
    }

    public static void Sum(double num1, double num2) {
        double ans;
        ans = num1 + num2;
        System.out.println(ans);
    }

    public static void Sum(int num1, int num2, int num3)
  {
        int ans;
        ans = num1 + num2 + num3;
        System.out.println(ans);
    }
  }
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac overloading.java } ; if ($?) { java overloading }
enrolment no :- 201130116009
Kartik patel
Sum of two integers
30
Sum of two double numbers
30.9
Sum of three integers
60
PS D:\kartik_java_prac>
```

AIM : Write a program to illustrate the concept of Single inheritance.

Code :-

```
class A {
    int a;

    void set_a(int i) {
        a = i;
```

```
    }

    void show_a() {
        System.out.println("The value of a=" + a);
    }
}

class B extends A {
    int b;

    void set_b(int i) {
        b = i;
    }

    void show_b() {
        System.out.println("The value of b=" + b);
    }

    void mul() {
        int c;
```

```
        c = a * b;
        System.out.println("The value of c= " + c);
    }
}

class inheritance {
    public static void main(String args[]) {
        System.out.println("enrolment no :-
201130116009");
        System.out.println("Kartik patel");
        A obj_A = new A();
        B obj_B = new B();
        obj_B.set_a(10);
        obj_B.set_b(20);
        obj_B.show_a();
        obj_B.show_b();
        obj_B.mul();
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac inheritance.java } ; if ($?) { java inheritance }
enrolment no :- 201130116009
Kartik patel
The value of a=10
The value of b=20
The value of c= 200
PS D:\kartik_java_prac>
```

AIM : Write a program to illustrate the concept of Multi level inheritance.

Code :-

```
class A {
    int a;

    void set_a(int i) {
        a = i;
    }

    void show_a() {
        System.out.println("The value of a= " + a);
    }
}

class B extends A {
```

```
    int b;

    void set_b(int i) {
        b = i;
    }

    void show_b() {
        System.out.println("The value of b =" + b);
    }
}

class C extends B {
    int c;

    void set_c(int i) {
        c = i;
    }

    void show_c() {
        System.out.println("The value of c =" + c);
```

```
    }

    void mul() {
        int ans;
        ans = a * b * c;
        System.out.println("The value of ans=" + ans);
    }
}

class mlinheritance {
    public static void main(String args[]) {
        System.out.println("enrolment no :-
201130116009");
        System.out.println("kartik patel");

        A obj_A = new A();
        B obj_B = new B();
        C obj_C = new C();
        obj_C.set_a(10);
        obj_C.set_b(20);
```

```
        obj_C.set_c(30);
        obj_C.show_a();
        obj_C.show_b();
        obj_C.show_c();
        obj_C.mul();
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac mlinheritance.java } ; if ($?) { java mlinheritance }
enrolment no :- 201130116009
kartik patel
The value of a= 10
The value of b =20
The value of c =30
The value of ans=6000
PS D:\kartik_java_prac>
```

EXPERIMENT NO : 6

Title :- How to use Exception handling

AIM : Write an exception class for a time of day that can accept only 24 hour representation of clock hours. Write a java program to input various formats of timings and throw suitable error messages.

Code :

```
import java.text.SimpleDateFormat;
import java.util.Date;

public class time {
    public static void main(String[] args) {
        Date dt = new Date();
        SimpleDateFormat dateFormat;
        dateFormat = new
SimpleDateFormat("kk:mm:ss");
        System.out.println("Enrollment :-
201130116009");
```

```
        System.out.println("Time in 24 hr format = " +
dateFormat.format(dt));
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac time.java } ; if ($?) { java time }
Enrollment :- 201130116009
Time in 24 hr format = 20:50:04
PS D:\kartik_java_prac>
```

EXPERIMENT NO : 7

Title :- : To Understand concept of abstract class .

AIM : The abstract Vegetable class has three subclasses named Potato, Brinjal and Tomato. Write an application that demonstrates how to establish this class hierarchy. Declare one instance variable of type String that indicates the color of a vegetable. Create and display instances of these objects. Override the toString() method of Object to return a string with the name of the vegetable and its color. .

Code :

```
public abstract class vegetable
{
 String color;
 public abstract String toString();
 public static void main(String a[])
 {
  System.out.println("enrollment no :- 201130116009
");
  System.out.println("Name :- Kartik patel");
  vegetable v=new Potato();
```

```
  System.out.println(v.toString());
  vegetable v1=new Brinjal();
  System.out.println(v1.toString());
  vegetable v2=new Tomato();
  System.out.println(v2);
 }
}
class Potato extends vegetable
{
 Potato()
 {
  color="white Potato";
 }
 public String toString()
 {
  return("Color of Potato: "+this.color);
 }
}
class Brinjal extends vegetable
{
```

```
 Brinjal()
 {
  color="dark purple";
 }
 public String toString()
 {
  return("Color of Brinjal: "+this.color);
 }
}
class Tomato extends vegetable
{
 Tomato()
 {
  color="Red";
 }
 public String toString()
 {
  return("Color of Tomato: "+this.color);
 }
}
```

OUTPUT :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac vegetable.java } ; if ($?) { java vegetable }
enrollment no :- 201130116009
Name :- Kartik patel
Color of Potato: white Potato
Color of Brinjal: dark purple
Color of Tomato: Red
PS D:\kartik_java_prac>
```

AIM : Create an abstract class 'Shape' with two members base and height , a member function for initialization and a pure virtual function to compute area(). Derive two specific classes 'Triangle' and 'Rectangle' which override the function area(). Use these classes in main function and display the areaof triangle and rectangle.

Code :

```
abstract class shape {
    int base, height;
    double a;

    void initFun() {
        base = 5;
        height = 8;
```

```
    }

    abstract void compute_area();
}

class triangle extends shape {
    public void compute_area() {
        a = (base * height) / 2;
        System.out.println("\n Area of triangle is " + a);
    }
}

class rectangle extends shape {
    public void compute_area() {
        a = (base * height);
        System.out.println("\n Area of Rectangle is " + a);
    }
}

public class shape1 {
```

```
    public static void main(String[] args) {
        System.out.println("Kartik patel_201130116009");
        triangle obj1 = new triangle();
        obj1.initFun();
        obj1.compute_area();
        rectangle obj2 = new rectangle();
        obj2.initFun();
        obj2.compute_area();
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac shape1.java } ; if ($?) { java shape1 }
Kartik patel_201130116009

 Area of triangle is 20.0

 Area of Rectangle is 40.0
PS D:\kartik_java_prac>
```

EXPERIMENT NO : 8

Title :- To Understand concept of interface.

AIM : Write a program that illustrate interface inheritance.Interface A is extended by A1 and A2.Interface A12 inherits from both A1 and A2.Each interface declares one constant and one method.Class B implements A12.Instantiate B and invoke each of its methods.Each method displays one of the constants.

Code :

```
interface A {
    final int a = 5;

    void dispP();
}

interface A1 extends A {
    final int a1 = 10;

    void dispP1();
```

```
}

interface A2 extends A {
    final int a2 = 20;

    void dispP2();
}

interface A12 extends A1, A2 {
    final int a12 = 12;

    void dispP12();
}

class B implements A12 {
    public void dispP() {
        System.out.println("dispP : " + a1);
    }

    public void dispP1() {
```

```
        System.out.println("dispP1 : " + a2);
    }

    public void dispP2() {
        System.out.println("dispP2 : " + a12);
    }

    public void dispP12() {
        System.out.println("dispP12 : " + a);
    }

}

class interfaceexa {
    public static void main(String arg[]) {
        System.out.println("Patel kartik_201130116009");
        B q = new B();
        q.dispP();
        q.dispP1();
        q.dispP2();
```

```
        q.dispP12();
    }
}
```

OUTPUT :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac interfaceexa.java } ; if ($?) { java interfaceexa }
Patel kartik_201130116009
dispP : 10
dispP1 : 20
dispP2 : 12
dispP12 : 5
PS D:\kartik_java_prac>
```

EXPERIMENT NO : 9

Title :- To Understand concept JavaFX and UI controls

AIM : Write a program that displays an image in a label, a title in a label, and text in the text area as follows:

Code :

```
package App;
import java.io.File;
import java.io.FileInputStream;
import javafx.application.Application;
import static javafx.application.Application.launch;
import javafx.scene.Scene;
import javafx.scene.image.Image;
import javafx.scene.image.ImageView;
import javafx.scene.layout.StackPane;
```

```
import javafx.scene.control.Label;



import javafx.stage.Stage;
public class App extends Application {
   public void start(Stage primaryStage) throws
Exception {
      StackPane root = new StackPane();
      FileInputStream fin = new
FileInputStream("D:/Manali/PicsArt_11-15-
08.19.32.jpg");
      Image img = new Image(fin);
      ImageView imgview = new ImageView(img);

      Label L = new Label("Vrajesh Patel", imgview);
      Scene scene = new Scene(root, 350, 350);
      root.getChildren().add(L);
      primaryStage.setTitle("Label Demo");
      primaryStage.setScene(scene);
      primaryStage.show();
   }
```

```
    public static void main(String[] args) {
        launch(args);
    }
}
```

Output :

PIC	NAME

EXPERIMENT NO : 10

Title :- To Understand concept of binary I/O, recursion and generics

AIM : Write a program to count the total number of chars, words, lines, alphabets, digits, white spaces in givenfile.

Code :

```
import java.io.*;
import java.util.*;

public class exc {
   public static void main(String[] args) throws
Exception {
      int char_count = 0;
      int word_count = 0;
      int line_count = 0;
      int wspace_count = 0;
      int digit_count = 0;
      int alphabet_count = 0;
      String fname;
```

```
        String temp_str;
        StringTokenizer token;
        BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));
        System.out.println("2011300116009_kartik
patel");
        System.out.println("Enter filename: ");
        fname = br.readLine();
        br = new BufferedReader(new FileReader(fname));
        while ((temp_str = br.readLine()) != null) {
            line_count++;
            for (int i = 0; i < temp_str.length(); i++) {
                if
(Character.isWhitespace(temp_str.charAt(i)))
                    wspace_count++;
                if (Character.isDigit(temp_str.charAt(i)))
                    digit_count++;
                if (Character.isLetter(temp_str.charAt(i)))
                    alphabet_count++;
            }
            token = new StringTokenizer(temp_str, " ");
```

```
            while (token.hasMoreTokens()) {
                word_count++;
                String s = token.nextToken();
                char_count += s.length();
            }
        }
        System.out.println("Character count: " +
    char_count);
        System.out.println("Word Count: " + word_count);
        System.out.println("Line Count: " + line_count);
        System.out.println("Alphabet Count: " +
    alphabet_count);
        System.out.println("Digit Count: " + digit_count);
        System.out.println("White Space Count: " +
    wspace_count);
        br.close();
    }
}
```

OUTPUT :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac exc.java } ; if ($?) { java exc }
2011300116009_kartik patel
Enter filename:
exc.java
Character count: 1160
Word Count: 146
Line Count: 45
Alphabet Count: 903
Digit Count: 20
White Space Count: 522
PS D:\kartik_java_prac>
```

AIM : Write the program for factorial using the concept of recursion.

Code :

```
import java.io.*;


class factorial {
    public int fact(int n) {
        int x, y;
        if (n < 0) {
            System.out.println("The Negative parameter in the factorial function.");
            return -1;
        }
        if (n == 0) {
            return 1;
        }
```

```
        x = n - 1;
        y = fact(x);
        return (n * y);
    }
}

public class excp {
    public static void main(String[] args) throws
IOException {
        factorial f = new factorial();
        System.out.println("201130116009_kartik patel");
        System.out.println("\nProgram for finding the
factorial of n.");
        System.out.print("\nEnter the number: ");
        BufferedReader buff = new BufferedReader(new
InputStreamReader(System.in));
        int n = Integer.parseInt(buff.readLine());
        System.out.println("\nThe Factorial of " + n + " is: "
+ f.fact(n));
    }
}
```

Output :

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac excp.java } ; if ($?) { java excp }
201130116009_kartik patel

Program for finding the factorial of n.

Enter the number: 8

The Factorial of 8 is: 40320
PS D:\kartik_java_prac>
```

```
PS D:\kartik_java_prac> cd "d:\kartik_java_prac\" ; if ($?) { javac kartik.java } ; if ($?) { java kartik }
201130116009_kartik patel
Sorted Array:-
A I K K R T
PS D:\kartik_java_prac>
```